

Ans:-  $T(x) = \sum_{i=1}^n \ln(1+x_i)$

Part B Q5  
(a)

Factorization Theorem  $\Rightarrow T(x)$  is a sufficient statistic of  $\theta$  if the likelihood function  $L(\theta|x)$  can be factored into 2 non-ve functions:-

$$L(\theta|x) = g(T(x); \theta) \cdot h(x)$$

where  $g$  depends on sample  $x$  only thro'  ~~$T(x)$~~   
&  $h$  does not depend on  $\theta$ .

\* Likelihood function:-  $L(\theta|x) \rightarrow$  product of individual densities  
$$L(\theta|x) = \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n \left[ \frac{\theta}{(1+x_i)^{\theta+1}} \right]$$

\* Factorization:- (separate terms involving  $\theta$ )

$$L(\theta|x) = \theta^n \prod_{i=1}^n \frac{1}{(1+x_i)^{\theta+1}} = \theta^n \prod_{i=1}^n (1+x_i)^{-(\theta+1)} = \theta^n \left[ \prod_{i=1}^n (1+x_i)^{-1} \right] \left[ \prod_{i=1}^n (1+x_i)^{-\theta} \right]$$

$$= \underbrace{\theta^n \exp\left(-\theta \sum_{i=1}^n \ln(1+x_i)\right)}_{g(T(x); \theta)} \cdot \underbrace{\prod_{i=1}^n \frac{1}{1+x_i}}_{h(x)}$$

depends on data  $x$  only thro' statistic  $T(x) = \sum_{i=1}^n \ln(1+x_i)$

depends on data  $x$  but not on parameter  $\theta$ .

$\therefore$  By Factorization Theorem the sufficient statistic for  $\theta$  is:

$$T(x) = \sum_{i=1}^n \ln(1+x_i)$$

equivalently any one-to-one transform function of  $T$  such as  $\prod_{i=1}^n (1+x_i)$  is also sufficient.

(85)

Part B (b)

To show minimal sufficiency, using Lehmann-Scheffé criterion:-

Statistic  $T$  is minimal & sufficient iff for any 2 samples  $x$  &  $y$   $\frac{L(\theta; x)}{L(\theta; y)}$  is free of  $\theta$  iff  $T(x) = T(y)$ .

From Part (a),

$$L(\theta; x) = \theta^n \prod_{i=1}^n (1+x_i)^{-1} \exp \left\{ -\theta \sum_{i=1}^n \ln(1+x_i) \right\}$$

$\therefore$  for  $x$  &  $y$ :

$$\frac{L(\theta; x)}{L(\theta; y)} = \frac{\prod_{i=1}^n (1+y_i)}{\prod_{i=1}^n (1+x_i)} \exp \left\{ -\theta \left( \sum_{i=1}^n \ln(1+x_i) - \sum_{i=1}^n \ln(1+y_i) \right) \right\}$$

The only  $\theta$  dependent factor is the exponential.

The ratio is constant in  $\theta$  iff.

$$\sum_{i=1}^n \ln(1+x_i) = \sum_{i=1}^n \ln(1+y_i)$$

ie., iff  $T(x) = T(y)$  where  $T(x) = \sum_{i=1}^n \ln(1+x_i)$

$\therefore T(x) = \sum_{i=1}^n \ln(1+x_i)$  is <sup>not only</sup> sufficient but ~~also~~ also minimal sufficient